THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL MCQ TEST SET 1

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS

50 × 1 = 50

Choose the correct option and write it in your answer book.

1. The LCM of 30, 42 and 66 is: 1

(a) 1260 **(b)** 1155 **(c)** 2310 **(d)** 6930

2. If α, β are zeroes of $x^2 - 4x + 3$, then $\alpha^3 + \beta^3$ is: 1

(a) 40 **(b)** 64 **(c)** 28 **(d)** 100

3. The pair $kx + 2y = k - 2$ and $8x + ky = k$ has no solution when $k =$ 1

(a) 2 **(b)** 4 **(c)** 8 **(d)** -4

4. If the roots of $x^2 - 7x + 10 = 0$ are α, β, then $\alpha^2 + \beta^2 =$ 1

(a) 39 **(b)** 29 **(c)** 49 **(d)** 69

5. If the 4th term of an AP is 9 and the 9th term is 4, then the 13th term is: 1

(a) 0 **(b)** 36 **(c)** 1 **(d)** 13

6. In $\triangle ABC$, $DE \parallel BC$ and $AD : DB = 1 : 2$. The ratio of areas of $\triangle ADE$ to trapezium $DBCE$ is 1: 1

(a) 9 **(b)** 4 **(c)** 8 **(d)** 10

7. The point on the y-axis equidistant from $(2, 3)$ and $(6, 5)$ has $y =$ 1

(a) 4 **(b)** 24 **(c)** 12 **(d)** -12

8. If $\sin\theta = \frac{4}{5}$ then $\sin^2\theta + \cos^2\theta + \tan^2\theta =$ 1

(a) $\frac{16}{9}$ **(b)** $\frac{16}{25}$ **(c)** $\frac{9}{25}$ **(d)** $\frac{25}{9}$

9. The angle of elevation of the top of a vertical tower from a point on level ground is 30°. After walking 4 m towards the foot of the tower along a straight line, the angle of elevation becomes 45°. The height of the tower is: 1

(a) $\frac{4\sqrt{3}}{3}$ m **(b)** $6 + 2\sqrt{3}$ m **(c)** $2 + 2\sqrt{3}$ m **(d)** 4 m

10. Tangents PA and PB are drawn from a point P, 50 cm from the centre O of a circle of radius 30 cm. The length of each tangent is 40 cm. The length of the chord AB is: 1

(a) 60 cm **(b)** 80 cm **(c)** 24 cm **(d)** 48 cm

11. A chord of a circle of radius 7 cm subtends an angle of 60° at the centre. (Use $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.73$; answer to two decimal places.) The area of the minor segment is: 1

(a) 21.19 cm² **(b)** 25.67 cm² **(c)** 4.47 cm² **(d)** 46.86 cm²

12. The largest possible solid sphere is carved out of a cube of edge 21 cm. The volume of the material wasted is: 1

(a) 4851 cm³ **(b)** 6835.5 cm³ **(c)** 9261 cm³ **(d)** 4410 cm³

13. The mean of a distribution with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60, 60–70, 70–80 and frequencies 3, 4, p, 4, 15, 10, 19, 22 respectively is 49. The value of p is: 1

(a) 25 **(b)** 23 **(c)** 28 **(d)** 22

14. Two fair dice are thrown together and the numbers on the upper faces are noted. The probability that the sum of the numbers is 8 is: 1

(a) $\frac{31}{36}$ **(b)** $\frac{1}{6}$ **(c)** $\frac{1}{9}$ **(d)** $\frac{5}{36}$

15. The least number that is divisible by every number from 1 to 6 is: 1

(a) 120 **(b)** 30 **(c)** 360 **(d)** 60

16. If α, β are zeroes of $x^2 - 5x + 6$, then $\alpha^2\beta + \alpha\beta^2$ is: 1

(a) 11 **(b)** 30 **(c)** 25 **(d)** 36

17. The pair $kx + y = k - 1$ and $4x + ky = k$ has infinitely many solutions when $k =$ 1

(a) 1 **(b)** 4 **(c)** -2 **(d)** 2

18. If the roots of $x^2 - 10x + k = 0$ differ by 2, then $k =$ 1

(a) 23 **(b)** 48 **(c)** 24 **(d)** 26

19. The sum of the first 13 terms of an AP is 390, and its first term is 6. The 13th term is: 1

(a) 66 **(b)** 60 **(c)** 54 **(d)** 24

20. In a right triangle with hypotenuse $c = 5$ and one leg $a = 3$, the area (sq. units) is: 1

(a) 6 **(b)** 10 **(c)** 12 **(d)** $\frac{15}{2}$

21. The area of the triangle formed by $(7, 0)$, $(0, 9)$ and the origin, doubled, is: 1

(a) 16 **(b)** $\frac{63}{2}$ **(c)** 63 **(d)** 126

22. If $\tan\theta = \frac{4}{3}$, then $\dfrac{1 - \tan^2\theta}{1 + \tan^2\theta} =$ 1

(a) $\frac{3}{4}$ **(b)** $-\frac{7}{25}$ **(c)** $\frac{7}{25}$ **(d)** $\frac{24}{25}$

23. A flagstaff stands on the top of a building 10 m high. From a point on level ground the angles of elevation of the bottom and the top of the flagstaff are 30° and 60° respectively. The length of the flagstaff is: 1

(a) 40 m **(b)** $20\sqrt{3}$ m **(c)** 20 m **(d)** 30 m

24. PA and PB are tangents from P to a circle, each of length 7 cm. A third tangent touches the circle at E on the minor arc AB and meets PA at C and PB at D. The perimeter of $\triangle PCD$ is: 1

(a) 7 cm **(b)** 14 cm **(c)** 21 cm **(d)** 9 cm

25. A brooch is a circular piece of silver wire of diameter 21 mm. The wire is also used for 4 equal diameters that divide the brooch into 8 sectors. The total length of wire required (in mm) is: 1

(a) 66 mm **(b)** 150 mm **(c)** 108 mm **(d)** 346.5 mm

26. A solid sphere of radius 10.5 cm fits exactly inside a cylinder (it touches the curved surface and both bases). The volume of the space in the cylinder not occupied by the sphere is: 1

(a) 7276.5 cm³ **(b)** 4851 cm³ **(c)** 2425.5 cm³ **(d)** 12127.5 cm³

27. The median of a distribution with classes 200–210, 210–220, 220–230, 230–240, 240–250, 250–260, 260–270 and frequencies 12, 8, x, 25, y, 18, 19 respectively, and total frequency 100, is 238.4. The value of x is: 1

(a) 11 **(b)** 9 **(c)** 8 **(d)** 10

28. Two fair dice are thrown together and the numbers on the upper faces are noted. The probability that the sum is less than 5 is: 1

(a) $\frac{5}{6}$ **(b)** $\frac{1}{6}$ **(c)** $\frac{7}{36}$ **(d)** $\frac{5}{36}$

29. The largest number dividing 67, 103 and 175 leaving remainder 4 in each case is: 1

(a) 18 **(b)** 9 **(c)** 1 **(d)** 3

30. If α, β are zeroes of $x^2 - 11x + 18$, then $|\alpha - \beta|$ is: 1

(a) 14 **(b)** 7 **(c)** 8 **(d)** 11

31. If $11x + 4y = 53$ and $5x + 3y = 30$, then $x + y =$ 1

(a) 5 **(b)** 8 **(c)** 3 **(d)** 15

32. For which value of k does $x^2 + kx + 9 = 0$ have equal roots (take $k > 0$)? 1

(a) 6 **(b)** 9 **(c)** 3 **(d)** 12

33. In an AP, $S_8 = 200$ and $S_7 = 154$. Then $a_8 =$ 1

(a) 46 **(b)** 52 **(c)** 40 **(d)** 354

34. The altitude on the hypotenuse of the right triangle with legs 3 and 4 is of length (cm): 1

(a) $\frac{7}{2}$ **(b)** $\frac{12}{5}$ **(c)** $\frac{5}{2}$ **(d)** 6

35. If $P(x, y)$ is equidistant from $(1, 3)$ and $(7, 5)$ and lies on $y = 0$, then $x =$ 1

(a) $-\frac{16}{3}$ **(b)** 4 **(c)** $\frac{16}{3}$ **(d)** $\frac{32}{3}$

36. If $\sin\theta = \frac{4}{5}$, the value of $2\sin\theta\cos\theta$ is: 1

(a) $\frac{24}{25}$ **(b)** $\frac{8}{5}$ **(c)** $\frac{12}{25}$ **(d)** $\frac{6}{5}$

37. A car moves at a uniform speed towards the foot of a vertical tower. The angle of elevation of the top of the tower changes from 30° to 60° in 4 minutes. The further time the car takes to reach the foot of the tower is: 1

(a) 8 minutes **(b)** 4 minutes **(c)** 2 minutes **(d)** $\frac{4}{3}$ minutes

38. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (15x + 4)$ cm and $QB = (8x + 25)$ cm. The length of QA is: 1

(a) 49 cm **(b)** 51 cm **(c)** 3 cm **(d)** 70 cm

39. A horse is tied at the midpoint of one side of a large square field with a rope 35 m long. The area of the field it can graze is: 1

(a) 110 m² **(b)** 55 m² **(c)** 1925 m² **(d)** 3850 m²

40. A solid cylinder of radius 6 cm and height 5 cm is melted and recast into a cone of the same base radius. The height of the cone is: 1

(a) 30 cm **(b)** 15 cm **(c)** 5 cm **(d)** 10 cm

41. **Assertion (A):** The HCF of 12 and 18 is 6 and their LCM is 36. 1
Reason (R): For two positive integers a and b, HCF(a, b) x LCM(a, b) = a x b.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

42. **Assertion (A):** A quadratic polynomial whose graph is a parabola cutting the x-axis at two distinct points has two real zeros. 1
Reason (R): The zeros of a polynomial $p(x)$ are the x-coordinates of the points where the graph of $y = p(x)$ meets the x-axis.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

43. **Assertion (A):** The graphs of $x + y = 4$ and $x - y = 2$ intersect at the point $(3, 1)$. 1
Reason (R): The coordinates of the point of intersection of the lines of a pair of linear equations give the solution of the pair.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

44. **Assertion (A):** The equation $2x^2 - 3x + 5 = 0$ has no real roots. 1
Reason (R): If the discriminant $b^2 - 4ac < 0$, a quadratic equation has no real roots.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

45. **Assertion (A):** The list $3, 7, 11, 15, \ldots$ forms an AP. 1
Reason (R): In an AP the difference between any two consecutive terms is the same constant.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

46. **Assertion (A):** In $\triangle ABC$, $DE \parallel BC$ with D on AB and E on AC. If $AD = 2$ cm, $DB = 3$ cm and $AE = 4$ cm, then $EC = 6$ cm. 1
Reason (R): A line parallel to one side of a triangle, meeting the other two sides at distinct points, divides those sides in the same ratio.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

47. **Assertion (A):** The distance of the point $(3, 4)$ from the origin is 5 units. 1
Reason (R): The distance of a point (x, y) from the origin is $\sqrt{x^2 + y^2}$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

48. **Assertion (A):** $\sin^2 60^\circ + \cos^2 60^\circ = 1$. 1
Reason (R): For every angle θ, $\sin^2\theta + \cos^2\theta = 1$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

49. **Assertion (A):** When an observer walks towards the foot of a tower, the angle of elevation of its top increases. 1
Reason (R): For a tower of fixed height h at horizontal distance d, $\tan\theta = \dfrac{h}{d}$, which increases as d decreases.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

50. **Assertion (A):** A tangent PT touches a circle with centre O at P. If $OP = 5$ cm and $PT = 12$ cm, then $OT = 13$ cm. 1
Reason (R): The tangent at any point of a circle is perpendicular to the radius through the point of contact.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

END OF QUESTION PAPER **Total: 50 Marks**